Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

Problem. Autonomous agents retrieve knowledge, interpret images, use tools, and reason through multistep plans. An unnoticed error can invalidate an otherwise plausible trajectory: a resupply agent may read the quantity from the wrong row of a photographed inventory sheet and pass it to a planning tool. Outcome rewards reveal only that the plan failed; generic self-critique may miss the original visual error. Reliable agents need to identify the faulty reasoning step and choose an appropriate correction.

Central hypothesis. We hypothesize that typed, evidence-linked self-assessment improves both inference-time recovery and learned reasoning policies. Agents that localize why a step fails should outperform outcome-only learning and scalar process rewards on unseen tasks at matched cost. A visual error calls for renewed observation; an invalid inference may call for backtracking. The diagnosis must improve the next action, not just label a mistake; unproductive retries should trigger abstention.

Research plan.

- **Ground each reasoning step in checkable evidence.** Represent intermediate claims and actions together with their source passages, knowledge-base facts, image regions, preconditions, and tool outputs. Build controlled multistep trajectories from entity and event relations, and connect natural agent traces to the evidence they used. The representation must distinguish a wrong observation from an invalid inference or tool action.
- **Learn typed, calibrated self-assessment.** Generalize our FG-PRM synthetic-error methodology from mathematical reasoning to agent trajectories: inject source misreads, entity mismatches, unsupported inferences, wrong tool arguments, and visual attribute or relation errors. Train verifiers to identify the earliest consequential error and its type. Test whether diagnoses trained on synthetic perturbations remain calibrated on natural failures, unfamiliar tools, and new visual inputs.
- **Turn diagnosis into agent self-improvement.** Condition the next action on the diagnosed failure: re-examine an image region, retrieve a missing fact, repair a tool argument, backtrack, or abstain. Use error-aware process feedback and successful corrective trajectories to train improved policies, while checking final outcomes independently to detect reward exploitation. Compare type-conditioned correction training against scalar process rewards, and test whether reducing one error type causes new failures elsewhere. Measure improvement across training rounds as well as immediate repair success, including cases where repeated self-correction wastes resources.

Evaluation and expected outcomes. Use MEQA for knowledge-grounded event reasoning, GAIA for multi-step tool and multimodal tasks, and τ-bench for state-changing tool interactions. Since these benchmarks do not provide complete typed error labels for intermediate steps, construct controlled error-injection variants with independently checked outcomes. Measure error localization, self-assessment calibration, successful repair, and task success per tool call. Evaluate diagnosis separately from recovery: detecting an injected error counts as useful only if the agent takes an appropriate next step, while final correctness must be checked independently. Compare to outcome-only RL, untyped process rewards, and LLM self-critique at matched cost.

Naval relevance. A Naval logistics support agent may combine a photographed inventory list, written guidance, and a resupply-planning tool. If it reads the wrong row, a useful self-assessment identifies the suspect image region and prompts a targeted correction before the tool result is trusted. This addresses the program's interests in uncertain reasoning and self-assessment, visual understanding, and the integration of knowledge, learning, and planning.

Prior work and PI qualifications. Our FG-PRM (Li et al., 2025, Findings of EMNLP) and HiPRAG (Wu et al., 2026, ICLR) develop process-level supervision for reasoning and retrieval. FaithScore (Jing et al., 2024, Findings of EMNLP) and FGAIF (Jing and Du, 2025, TMLR) provide starting points for multimodal feedback. The PI holds a 2024 NSF CAREER award, received a 2023 Amazon Research Award for process-guided fine-tuning, and, as a postdoc, led system integration for the RESIN team's RESIN-11 delivery system in DARPA KAIROS.